

CORNERSTONE Open Platform — OPENSOURCE PDK live monitor

User Manual — v1.0.0 (rev. 3)

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1. What this dashboard is

The **CORNERSTONE Open Platform PDK Monitor** is a single-file HTML dashboard for tracking, sharing, and collaborating on silicon-photonics Process Design Kit (PDK) content. Open `pdk-monitor.html` in any modern browser — no install, no server, no build step. All state lives locally in the browser's IndexedDB and localStorage; sharing happens through GitHub.

The dashboard organises PDK content as:

- **Technology platforms** (SOI 220 nm, TFLN 300 nm, SiN 300 nm, etc.) at the top level — one column per platform.
- **Building Blocks (BBs)** inside a platform (waveguides, MMIs, grating couplers, photodetectors, ...) grouped by BB type.
- **Per-BB tabs**: Overview · Layout (GDS) · S-matrix · Tests · Fab · Scripts · Documents · Variants.
- **GDS Layers** dictionary with name, layer/datatype, description, colour.
- **User-admin pool** — multiple contributors with their own users/<id>/ folder on the repo; the admin merges all slices.

2. Quick start

1. Download `pdk-monitor.html` from the v1.0.0 release.
2. Open it in any modern browser. The dashboard boots into **Read-only** mode with the factory defaults.
3. Click the mode pill in the header. The default admin password is `CS2026_PDK_Design&Test` — change it from *Manage* → *Passwords* after first sign-in.
4. To pull a real PDK, open a Platform column → *Edit* → paste the GitHub folder URL → save → ↓ *Import platform*.

3. Modes & sign-in

Mode	Capabilities
Read-only	View everything, run scripts, search & download, resize columns + side panel, clear local cache
User-admin	All Read-only + add / edit / remove items in their own users/<id>/ slice
Admin	Full control across the whole dashboard

4. Header bar

- **Mode pill** — sign in / out (default password CS2026_PDK_Design&Test).
- **Search** — full-text across BBs / files / documents / datasets.
- **User filter** — narrow visible BBs by publisher.
- **↓ GitHub fetch (all)** — admin global fetch (components/sparams/... convention).
- **Distribution** — opens the Distribution modal.
- **Reset / Clear cache / Reset my changes** — context-sensitive per role (see § 12).

5. Platforms tab

Each column is a platform. BBs inside a platform are grouped by BB type. The per-platform group head has a BB-type shortlist filter.

5.1 Importing a platform from GitHub

1. Click **Edit** on a platform column.
2. Set the GitHub folder URL.
3. Save → click **↓ Import platform**.

Each .gds under components/ becomes a BB. Layer / datatype info is read from the GDS header and propagated to the BB.

6. Per-BB tabs

Click a BB card to open the side panel:

- **Overview** — metadata + custom sketch upload.
- **Layout** — GDS preview, raw download, **auto-generated YAML attached as a Document on upload** (see § 7).
- **S-matrix** — folders for .csv / .xlsx / .s2p-.s8p / .json / .tsv / .dat. Plot parsed traces directly.
- **Tests** — same parsers as S-matrix.
- **Fab** — fabrication artefacts.

- **Scripts** — Python scripts run in-browser (Pyodide sandbox).
- **Documents** — datasheets, YAMLs, configs (up to 50 MB each).
- **Variants** — alternate copies attached as subBlocks.

6.1 Panel footer

The side-panel footer carries the BB-level actions:

- **Delete block** (owner / admin only) — removes the BB.
- **Download .zip** (*new, rev. 2*) — packs every file on this BB into a single zip with the CORNERSTONE folder layout (see § 8). Available in every mode.
- **Close, ↺ Sync (N), Edit** (right cluster).

7. Auto-generated YAML on GDS upload (*new, rev. 2*)

When you upload a .gds for a BB:

1. The GDS is parsed; the layers actually used in the rendered cell are recorded.
2. A YAML descriptor is auto-built and attached to the BB's Documents tab as <bb-slug>.yaml. Re-uploading the GDS refreshes the YAML (replaces the previous auto-YAML).
3. You can edit the YAML afterwards like any other document.

Sample auto-YAML (format mirrors the cornerstone-community/Si_220nm_passive convention):

```
name: MMI_1x2
description: ""
bb_type: mmi
category: passive
platform: SOI 220 nm
status: available
owner: john-smith
gds_file: components/MMI_1x2.gds
```

layers:

- layer: 3
 - datatype: 0
 - name: Si_220nm_pass_etch
 - description: 220 nm Si etch
- layer: 100
 - datatype: 0
 - name: SiO2_220nm_pass

parameters:

```
L_taper: 12.5
W_taper: 2.0
```

Layer entries are auto-correlated against MODEL.layers so the name / description come from your global layer dictionary.

8. BB-level zip download (new, rev. 2)

Click **Download .zip** in the BB panel footer to pack every file on the BB into a single zip. The zip uses the CORNERSTONE convention (mirrors cornerstone-uos/cornerstone-community):

```
<bb-slug>/
components/
  <bb-slug>.gds          ← raw GDS bytes
  <bb-slug>.yaml         ← auto-generated metadata
sparams/<bb-slug>/
  *.csv | *.s2p | *.s4p ...
characterisation/<bb-slug>/
  *.csv | ...           ← test results
fabrication/<bb-slug>/
  *.png | ...
scripts/
  *.py
documents/
  *.pdf | *.md | ...
files/
  *
README.txt              ← manifest with counts + paths
```

Available in every mode — it's a read-only export.

9. GDS Layers list

The top-level Layers tab carries the layer dictionary. Each row: layer / datatype, human name, description, colour swatch. Hoverable; sort by layer #; filter by BB type via the Manage-list dropdown.

10. Distribution — multi-user pool

Open **Distribution** from the top bar.

10.1 Section 1 — Shared repositories

Add sources ({ repo URL, branch, path }). Enable the ones you want to fetch from.

10.2 Section 2 — Fetch & adopt

Primary action row:

- **Fetch all enabled sources** — pulls the configured dashboard.json from every enabled source.
- **Discover user slices** — explicit multi-pick walker.
- **Fetch my slice** (*user-admin only*) — pulls / bootstraps the user's own slice.
- **Merge into my dashboard** — additive fold of every staged payload.
- **Replace whole dashboard** (*danger*) — wipe local copy + use only the fetched payload. A Backup .json downloads first.

⚠ Important (rev. 2)

Below the action row there's a checkbox:

- ☐ Also auto-pull user slices when fetching (off by default — main repo only)
- **Default OFF** — Fetch all enabled sources now pulls ONLY the configured main dashboard.json. Clean admin snapshot, no silent fold-in of every user-admin contribution.
- **Tick the box** to restore the prior auto-walk of every plausible users/ location.
- Discover and Fetch user repos buttons always walk users/ regardless of this checkbox.

Advanced drawer — selective pool fetch (rev. 2)

Open ⚙ *Multi-user pool sync (advanced)*. Click **Fetch user repos** — instead of pulling everything immediately, the dashboard now shows a **checklist** of every discovered slice:

☐ Selective fetch — 6 user slices discovered
☒ Select all 6 / 6 ticked
☐ Fetch & install 6 selected] [Cancel]

☒ john-smith cornerstone-uos/PDKMonitor
 users/john-smith/dashboard.json
☒ sara-park ...
 ...

Tick which slices to install, click **Fetch & install N selected**, and only the ticked slices are fetched + hydrated + staged for merge.

10.3 Section 3 — Publish

- **Publish as Pull Request** — review-before-merge.
- ⚡ **Direct commit** — straight to the branch. For user-admins: auto-routes to users/<your-id>/dashboard.json and externalizes binaries to real files under users/<id>/<section>/<bb>/<file>.
- **Seed user folders from sign-in list** (*admin*).
- **Wipe my repo folder** (*user-admin, danger*).

10.4 Section 4 — Duplicate resolver

Find BBs / groups / tabs sharing a name. Two kinds: **same-parent duplicates** (same platform) and **cross-platform overlaps** (same name across different platforms).

Bulk select + bulk actions

Toolbar at the top:

Found 7 clusters: 2 BB · 0 platform · 0 tab · 5 cross-platform

☐ Select all N selected

☐ Merge N selected] ☐ Variants N selected] ☐ Auto-resolve all (7)]

Every cluster has a checkbox. Tick the ones to act on, then use the toolbar buttons. **Auto-resolve all** includes cross-platform clusters too (was missing in earlier builds).

Per-row Skip / Merge-in (rev. 2)

For each non-keeper row in a cross-platform cluster:

[☐ Keeper?] [☒ Merge in] MMI_1x2 — bb_abc1 in SOI 220 → Passive
[☒ Keeper?] [☒ ★ Keeper] MMI_1x2 — bb_def2 in TFLN 300 → Linear
[☐ Keeper?] [☐ Skip] MMI_1x2 — bb_ghi3 in SiN 300 → Active

- Tick “Merge in” (default) → that copy folds into the keeper.
- Untick → **Skip** — that copy is left alone.

11. Real-files architecture

User-admin publishes write each binary item as a separate file under:

users/<your-id>/<section>/<bb-name>__<id-suffix>/<filename>

The dashboard.json carries metadata + repoPath references only. Pool members hydrate the real bytes on fetch.

Add → upload locally + ↗ Direct commit. **Remove** → delete locally + ↗ Direct commit (orphan auto-prune). **Nuke** → Wipe my repo folder.

12. Reset button — role-aware (rev. 2)

The Reset button in the header is now visible in every mode and adapts to the role:

Role	Label	What it does
Read-only	Clear cache	Wipes IndexedDB + localStorage + sessionStorage; reseeds defaults. Nothing on GitHub

User-admin	Reset my changes	is touched. Removes only the user's own contributions.
Admin	Reset	Wipes everything + reseeds factory defaults.

13. Layout configuration in read-only (rev. 2)

Read-only users can now configure their visual dashboard layout:

- **Column width** — drag the right edge of any platform column.
- **Side panel width** — drag the left edge of the BB detail panel. Width persists in localStorage so it survives a refresh.

Both are pure local-state changes — no GitHub interaction.

14. Ownership & per-item delete (rev. 2)

canModifyId(id) now also accepts the per-item who field. On every model load, ensureLists backfills MODEL.ownership[id] from item.who if the map entry is missing. Fixes the long-standing “user-admin can’t delete their own legacy uploads” issue: open v1.0.0 with an existing dashboard, your prior uploads light up with the proper delete button.

15. Sketches — procedural BB previews

Every BB without a custom sketch gets an auto-generated material-aware SVG preview (waveguide, MMI, MZI, ring, DC, bend, crossing, grating coupler, edge coupler, spiral, **photodetector**, modulator). All sketches use the platform’s own colour from matStyle. The photodetector is **material-neutral** (no Ge baked in — the absorber gradient is derived from the platform’s material colour).

16. Search and filter

- Global search across BBs / files / documents / datasets / scripts.
- User filter (multi-pick).
- BB-type shortlist per platform.

17. Local scripts

Python (Pyodide). Globals: data (S-matrix datasets), tests, bb. **Run locally...** packs script + every BB file into a folder with VS Code / PyCharm / Sublime deeplinks.

18. Backups

- Auto on Replace, on every resolver action, before publish prune.
- Manual export via *Manage* → *Export local state*.

19. Roles and security

Item

Admin password

User-admin sign-ins

GitHub token

20. Troubleshooting

Symptom	Fix
Fetch finds no user folders	Walker tries sibling, parent, repo-root, and OpenSource_PDKMonitor/users automatically. Confirm the source is enabled.
Auto-resolve says 0	v1.0.0 rev. 2 includes cross-platform clusters. Click Auto-resolve all again.
User-admin can't delete own file	rev. 2 backfills ownership from who on load. Re-open the dashboard.
pulls user slices unexpectedly	Untick “Also auto-pull user slices when fetching” — main repo only.
Local cache too full	Read-only: click Clear cache.

21. Licence

TAPR Open Hardware License v1.0 for hardware design content distributed through the dashboard. See LICENSE-TAPR.txt.

22. Credits

Emre Kaplan — a.emrekaplan@gmail.com CORNERSTONE Open Platform — <https://www.cornerstone.sotonfab.co.uk/>

23. Revision 3 — Prepare-for-PDK, cooldowns, fetch fidelity

Rev. 3 keeps the v1.0.0 stamp. The headline items below all land in the same bundle as the v1.0.0 release; this section explains them in order.

23.1 Prepare-for-PDK — per-section file multi-select

Every row of the Prepare-for-PDK section-target table now has a  **File**. Expand to see every file that section would commit, each with its own checkbox. Select-all / Clear buttons sit at the panel header.

Default — all files ticked. Untick individuals to exclude them (e.g. ship the GDS + YAML, leave WIP datasets behind; or skip one noisy test file).

Selection persists per-BB under `localStorage["pdk-monitor.pdkPrepTargets"]` — re-opening the modal remembers your last choice.

Switching the section dropdown clears that row's selection because the new section has a different file catalogue.

Unticked items are reported in the log ( ... (unticked in file picker)) and counted in the Summary line.

23.2 Dual publish buttons with cooldowns

The Prepare-for-PDK modal now shows two publish buttons side-by-side.

Open Pull Request — groups files by target repo (one PR per owner/repo/branch). Throttled to 1 PR per hour per signed-in identity to keep reviewers' notifications sane. Failed PRs don't burn the slot.

✂ Direct commit — commits straight to each target branch. Lenient progressive backoff: 0 s, 30 s, 90 s, 180 s, 300 s, 600 s, 1200 s. Allows 2 commits inside 1 min and 5 commits inside 10 min, then gradually slows.

Both buttons show a 1-Hz live countdown ( **PR available in 27m 03s, 14s**) so the user always knows when the next attempt opens.

Admin role bypasses both. State is per-identity and persisted in `localStorage` so multiple users on the same machine don't share quota.

23.3 No README / no SVG on Prepare-for-PDK publish

The auto-generated `<bb>.README.md` upload has been removed from the commit flow. Only principal files (GDS / YAML / datasets / scripts / docs / `eda_links.json` / variants) land in the community PDK target. When publishing as a PR, the PR body still carries the description — reviewers see context, the repo stays clean.

SVG previews are no longer uploaded either. They're procedural and regenerable from the GDS or BB type at render time; shipping them was clutter.

23.4 Editable target URL — Commit tests to repo

The previous static target URL on the Commit-tests-to-repo modal is now an editable input pre-filled with the computed `<platform>/test/<bb>/` path. Edit it freely to redirect tests anywhere (e.g. your own users/`<id>/<bb>/tests/`). A Reset to default button repopulates the field. Saved per-BB in `localStorage["pdk-monitor.commitTestsTarget"]`.

23.5 Scripts + S-matrix GitHub fetch — every file type

Scripts fetch (↓ GitHub fetch on the Scripts tab) used to accept only .py files; everything else silently went to Tests. Rev. 3 routes:

Textual extensions

(.py, .ipynb, .sh, .bash, .js, .ts, .m, .r, .jl, .f90, .c/.h/.cpp, .java, .go, .rs, .sql, .md, config formats) → Scripts as editable source.

Binary files (.exe, .so, .jar, ...) → Scripts as base64 binaryContent — same shape as the Upload binary button. Browsers still can't execute them directly; use Run locally to bundle them with a terminal.

S-matrix fetch now accepts touchstone variants (.s/y/z/h/gNp), HDF5/.h5, .tdms, .mat, .npz/.npy, .prn, .asc, .out, .log, .pickle/.pkl, .cti/.cit, plus the previous CSV/TSV/XLSX/JSON/TXT set. Small extension-less files are kept (some lab tools emit untyped touchstone). README / license / config noise is still dropped.

23.6 Lazy-hydrate on Download — cross-user files

When a file card carries a repoPath but no in-memory bytes (the typical state for files contributed by another user-admin), clicking ↓ Download now walks every enabled Distribution source, fetches the bytes via the GitHub Contents API (with git/blobs fallback for files > 1 MB), writes them back onto the record, and completes the download with the original filename and bytes.

First click hits GitHub; subsequent clicks are local.

The saved Blob is now tagged with the correct Content-Type (text/csv, application/pdf, image/png, application/x-yaml, etc.) instead of generic application/octet-stream — the OS picks the right icon / app.

If a file truly has no repoPath and no in-memory bytes, the sidecar JSON is still written, but the message now points at the right fix (configure a Distribution source pointing at the publisher's repo).

23.7 Manage-list BB types — proper label everywhere

Adding a new BB type via Manage lists mints it with an id like x_9pogr1s. That id was leaking onto the BB Details page's Type field, the auto-YAML, the PR body, the markdown portfolio report, and the Excel export. Rev. 3 introduces bbTypeLabel(t), a resolver (same shape as statusLabel / categoryLabel) that finds the matching MODEL.bbTypes entry by id or name and returns the human label. Plumbed through every user-facing render site.

The global search index now matches against both the raw id and the resolved label, so typing "heater" finds BBs whose bbType id is x_9pogr1s but whose Manage-list label is "heater".

23.8 BB-tabs scrolling

Three nested scroll containers used to fight wheel events inside Tests / S-matrix / Fab tabs (`.panel-body`, `.folder-list-wrap`, `.folder-body`). The unreachable- $\times$ bug and the "scroll doesn't work at all" regression had the same root cause. Rev. 3 untangles it:

`.panel-body` and `.modal-body` get `min-height: 0` so the flex layout can shrink them to their allocated space and the outer scrollbar engages.

`.folder-list-wrap` drops its viewport-relative max-height cap — the outer panel scroll covers it end-to-end.

`.folder-body` keeps an inner scroll at `max-height: 280 px` (~3 file cards) so a 30-file folder stays compact, with overscroll-behavior: contain so wheel events fall through to the outer scroll once the inner reaches a boundary.

Extra padding-bottom + scroll-padding-bottom so the last row's wrapped action buttons (including $\times$) are always reachable.

23.9 User-admin edit rights

`canModifyId(id)` is now: `admin` $\rightarrow$ `true`, `user-admin` $\rightarrow$ `true` regardless of who created the item, `read-only` $\rightarrow$ `false`. This switches the user-admin role to a collective contributor model — every user-admin can edit any BB, not just the ones they personally created. Each item still carries a `who` field so attribution stays accurate on publish.

23.10 Layout section accepts GDS + OASIS

The Layout tab's upload button is now labelled Upload `.gds` / `.oas` / `.svg`. The file picker accepts `.gds`, `.oas`, `.oasis`, and `.svg`; the upload-mode tab reads Layout (GDS/OAS/SVG). GDS and OASIS are the essential layout formats — SVG remains as a preview-only fallback.

`.gds` — parsed in-browser; polygons render coloured by layer (legend below the layout). Auto-YAML still attaches a descriptor to the BB's Documents tab on upload (chapter 7).

`.oas` / `.oasis` — stored verbatim under the same layout slot as GDS. Download `.oas` appears alongside Download `.gds`. The viz stage shows a placeholder card with the filename, size, and a hint to open the file in KLayout (free, klayout.de) for the full polygon view — no in-browser OASIS parser is bundled.

`.svg` — renders directly. Useful as a preview when the foundry hasn't shipped a GDS or OAS for the BB yet.

↓ GitHub fetch picks `.gds`, `.oas`, and `.oasis` when given a folder URL — preferring a name-matched GDS first, then a name-matched OAS, then any GDS, then any OAS. Paste a direct file URL to skip the auto-pick.

Prepare-for-PDK's per-section file picker labels OAS entries as (OASIS layout) and GDS entries as (GDS layout) so they're unambiguous when multi-selecting which files commit to the community PDK.

Round-trip fidelity — OAS bytes survive ↓ Download, ⚡ Direct commit, ♻ Sync exactly as uploaded; KLayout consumers see the original file.

Pull Request,

24. Revision 4 — admin safety, cross-check, variants, polish

Revision 4 sits on top of the same v1.0.0 stamp. It hardens the multi-user distribution model, adds a full cross-dashboard cross-check analysis suite with three pie charts and selective Excel/zip export, surfaces variants as their own category in the UI and the analysis, opens up admin-editable footer texts, fixes the file-list scroll across every BB tab, and applies several permission and layout polishes.

24.1 CRITICAL — admin safety gate against publishing into users/

Three layers of defence now stop an admin's Direct commit from wiping every contributor's folder under users/:

The Direct-commit confirmation flow hard-blocks any admin push whose normalised path starts with users/ — toast + alert + log entry, no API calls fired.

Externalize fires only when the path matches the strict pattern users/<id>/dashboard.json — i.e. a per-user slice. The root users/dashboard.json no longer triggers externalize.

pruneOrphanRepoFiles refuses any usersDir value that isn't users/<exactly-one-segment> and returns a no-op summary, so even if a caller mis-routes, the contents of users/ cannot be deleted.

If a previous publish has already mutated the users/ folder, open the repo's Commits view on GitHub, find the offending commit, click Revert this commit, or run ``git checkout <pre-incident-sha> -- users/'` locally.

24.2 Cross-check analysis vs. another dashboard

A new Cross-check vs. uploaded dashboard button sits next to the Resolve duplicates button. Click it, pick another dashboard JSON file (full snapshot, slice, exported cross-check payload, or even a flat entities[] / blocks[] / platforms[] list) and the dashboard:

Walks every BB across both sides, recursing into subBlocks / variants / children.

Matches by (lowercased name) @@ (lowercased platform).

Splits the result into Common / Here-only / There-only.

Renders three SVG donut charts: per-platform counts, here/there/common, and variants as a separate category so a contributor's variant additions are visible at a glance.

Shows a side-by-side summary card with w/o variants and with variants counts per side.

Exposes a Download Excel report button — multi-sheet (Summary, Common, Here-only, There-only, Variants here, Variants there, Per-platform counts).

Per-BB checkboxes on the Here-only / There-only lists with a Download selected (.zip) action — bundles every file associated with each ticked BB (GDS / OAS / S-matrix / tests / fab / scripts / documents / EDA links / files) into a single zip with the canonical CORNERSTONE folder layout.

24.3 Variants as a first-class category

A recursive walker flags every sub-block / variant with `isVariant=true` and a `parentName` link. The flag flows into the cross-check donuts, the Excel report, and the BB-list header bar so the user can see at a glance how many variants a dashboard carries. The main header now shows two new chips: `N variants` and `M total`. The tooltip on `M total` clarifies that the `X blocks` count to the left excludes variants, while `M total` is top-level BBs + every recursive variant.

24.4 Admin-editable footer

Admin mode now exposes inline editors for the three footer strings — Terms link text, the hardware-licence sentence (TAPR Open Hardware License), and the software-licence sentence. Edits persist to `localStorage` under `pdk-monitor.legalTexts`, with a `reset-to-default` button per field. Public read-only renders the saved text verbatim.

24.5 ? Clear cache in every role

The Clear cache button is now exposed in admin, user-admin, and read-only modes with the same effect: wipe `IndexedDB` + `localStorage` + `sessionStorage` and reseed factory defaults. Nothing on GitHub is touched. The duplicate Clear cache button that appeared next to it in read-only after the role-parity work has been collapsed back to a single button.

24.6 Prepare-for-PDK opened to Admin

The Prepare for community PDK action no longer hides behind user-admin gating. Admins can now use the same modal to push curated files into the community repo. Admin Direct commit still bypasses cooldowns and is still blocked from publishing into users/ by the safety gate described in 24.1.

24.7 Scroll fixed across every BB tab

Three nested scrollers (`.panel-body`, `.folder-list-wrap`, `.folder-body`) had been fighting over the same wheel events. Fix: `.panel-body` and `.modal-body` get `min-height: 0` so they shrink inside their flex parent and engage their own `overflow-y: auto`. `.folder-list-wrap` drops its 72vh cap. `.folder-body` keeps an inner ~3-item scroll at `max-height: 280px` with `overscroll-behavior: contain` and extra `scroll-padding-bottom` so the `×` on the last row always reaches the viewport.

24.8 Tab-named externalize subfolders

Per-user externalize now writes binaries under tab-named subfolders instead of the legacy `PDKMonitor_*` prefix:

`users/<id>/<bb-slug>/<section>/<filename>`

where `<section>` is one of `layout`, `smatrix`, `tests`, `fab`, `eda`, `scripts`, `documents`, `files`, `variants`. Existing items still load from the legacy paths; new commits use the new layout. EDA links serialise to `<bb-slug>/eda/eda_links.json` on publish so adopters fetching the slice see the EDA / cloud-IDE references too.

24.9 Instant BB-card refresh after upload

A debounced `scheduleBoardRefresh()` hook in `saveModel` re-renders the board 40 ms after every model save, so BB card badges (file counts, tab indicators, status pills, the new variant chip) update immediately after an upload without waiting for the next interaction.

24.10 Platform column min-width — content-aware

Columns can now shrink until the technology platform name is about to be covered by the Sync / Import buttons, then stop. `.title-area` uses flex: 1 0 auto so action buttons wrap onto the next line before the title shrinks. Group-head also wraps so BB-type categories flow gracefully instead of forcing horizontal scroll. Each BB row reserves enough min-width that at the narrowest column setting the first 5 letters of the BB name remain visible.

24.11 Custom BB type labels everywhere

Custom BB types added via Manage-lists get a `uid("x")` id (e.g. `x_9pogrls`). A new `bbTypeLabel(t)` resolver looks up the matching `MODEL.bbTypes` entry by id or name and is plumbed through every user-facing render site — BB Details Type field, YAML export, PR body, markdown portfolio report, Excel export, search index. Typing "heater" in search now finds BBs whose `bbType` is `x_9pogrls` but whose Manage-list label is "heater".

24.12 Permissions — collective contributor model

User-admins can now edit BBs created by other users. `canModifyId(id)` is `admin → true`, `user-admin → true` regardless of owner, `read-only → false`. Destructive operations (delete BB, delete file) still require ownership; everyone can add files and edit details on any BB.

24.13 Lazy-hydrate downloads + correct MIME

When a file card carries a `repoPath` but no in-memory bytes — the typical state for files contributed by another user-admin after the publish-strip — clicking  Download now walks every enabled Distribution source, fetches the bytes via the GitHub Contents API (with `git/blobs` fallback for files > 1 MB), writes them back onto the record, and completes the download with the original filename + bytes. First click hits GitHub, every subsequent click is local. The saved Blob is tagged with the right Content-Type (`text/csv`, `application/pdf`, `image/png`, etc.) so the OS opens it with the right app.

25. Revision 5 — AI Insight (per-BB summaries, chat agent, contributor Excel)

Revision 5 adds an end-to-end AI synthesis layer on top of the existing dashboard. The implementation is hybrid: deterministic metric extraction runs in the browser, and a configurable LLM only narrates the numbers — every quoted value is checked against a citation map and any number the model invented gets a wavy red underline. The release also ships a conversational chat agent with read-only tool use, a contributor Excel export, and a modern colour-ring 'AI' logo on every entry-point.

25.1 Configuring the AI endpoint

Click the new gradient AI Insight button in the header (the colour-ring badge), then click Configure endpoint, model, API key, monthly cap. Fields:

Endpoint — defaults to `https://api.anthropic.com/v1/messages`. Works with Anthropic, OpenAI (`https://api.openai.com/v1/chat/completions`), OpenRouter, Ollama / vLLM / LM Studio (`http://localhost:11434/v1/chat/completions`), or your foundry's own OpenAI-compatible gateway. The API family is auto-detected from the URL.

Model — model alias supported by the chosen endpoint. Default 'claude-sonnet-4-5'. For OpenAI use 'gpt-4o-mini' / 'gpt-4o'. For OpenRouter prefix with the vendor: 'anthropic/claude-sonnet-4.5'.

API key — paste the raw secret. Stored in localStorage on this browser only; never published. Two helper buttons sit next to the field: reveals the value, copies it to the clipboard.

Monthly token cap — input + output tokens, summed monthly. When the cap is reached every button disables until you raise it or the calendar month rolls over.

Egress mode — three levels: metrics-only (safest; sends only the computed JSON), metrics+extracts (also includes ≤ 4 KB extracts from YAML / Markdown), full (also includes ≤ 50 KB file payloads).

Prompt style — brief / balanced / deep. Deep asks the model for an explicit fabrication-tolerance interpretation when cohort statistics suggest one.

A live meta line under the key field shows length, prefix-detected family (Anthropic sk-ant-..., OpenAI sk-proj-..., etc.), and a head...tail preview. Two warnings fire automatically: leading/trailing whitespace, and endpoint/key family mismatch. If you're seeing 401s, click

Test connection — it fires a minimal call against the form's current values (not the saved settings) and reports back HTTP code, response time, tokens used, and a targeted hint for each common failure mode (401 / 403 / 404 / 429 / 5xx / CORS).

25.2 Per-BB AI tab

Every BB Details panel now has a new AI Insight tab. Opening it shows three things even before any API call:

File-inventory chip row — counts of S-matrix datasets, test files, scripts, documents, fab folders, variants, plus a green chip if a GDS/OAS layout is present.

Deterministic headline metric cards — S21 peak, S31 peak, |S21-S31| imbalance, S21 1-dB bandwidth, S11 / S22 reflections. Each card shows mean $\pm \sigma$ with n. These numbers come from a pure-JS walker over every parsed dataset and test file. They're correct even with no LLM call.

Cached AI summary (if one exists) — a banner shows cached, timestamp, model, and tokens; the prose appears below with citation tags hyperlinked to the citation map drawer.

Click `Summarise this BB` to send the metric JSON + citation map to the configured LLM and render its response. Citation discipline: every numeric token in the response is matched against the citation map at 1-, 2-, and 3-decimal precision; tokens that don't match get a wavy red underline and the tooltip 'Number not found in the citation map — may be hallucinated'. Open the `Citation map` drawer at the bottom of the tab to see every value the model was allowed to quote, with source + scope + unit.

Caching is content-keyed by SHA-256 of (metrics JSON + model + prompt style). Re-clicking on unchanged data returns in under 100 ms with zero token cost. Each cache entry stores model + endpoint + egress mode + tokens + created_at + created_by + extractor version (provenance for audit).

Bottom row of actions: `Refresh summary` (re-runs, costs tokens), `Copy summary` (clipboard), `Download .md` (file), `Settings` (jump to AI Settings), `Clear cached summary` (this BB only).

25.3 Platform + dashboard executive summaries

From the main AI Insight panel:

Per-platform — click the `Brief` button next to any platform name in the Per-platform shortcuts list. The dashboard loops every BB on that platform, takes the one-paragraph headline from each cached summary (computes missing ones), composes a single Markdown brief, and downloads it as `<platform>-executive-summary.md`.

Whole dashboard — the `Executive summary — whole dashboard (.md)` action does the same for every platform and emits one `dashboard-executive-summary.md` you can paste into a shuttle report.

Both summaries are computed from cached headlines where available, so a fresh full roll-up across 30 BBs costs around a single Sonnet-class call's worth of tokens — not 30 individual calls — provided the per-BB cache was populated first.

25.4 Chat agent — multi-turn, tool use

Click `Chat with the agent` in the main AI panel. The agent has six read-only tools that walk the dashboard MODEL:

`dashboard_overview()` — platforms / BBs / variants / contributors / file counts.

`list_platforms()` — every Technology Platform with material, thickness, BB count, GitHub URL.

`list_bbs(platform?, bb_type?, status?, owner?, search?)` — filtered listing with file-inventory snapshots.

`get_bb_details(name, platform?)` — full metric block for one BB (same shape as the AI tab).

`search_dashboard(query, limit?)` — free-text search across names, descriptions, owners, parameters.

`get_contributor_stats(contributor?)` — per-contributor activity sorted by total contributions.

`summarise_bb(name, platform?, force?)` — runs or returns cached AI summary for one BB.

Type a question and press Send (or Cmd/Ctrl+Enter). The agent decides which tools to call, the dashboard executes them locally, returns JSON, and the agent either calls more tools or writes the final answer. Every tool call is shown inline with its arguments; every result lands in a collapsible details panel so you can see the raw data the agent read. Hard cap of six tool-use iterations per turn protects against runaway loops.

Conversation history persists per browser in `localStorage` (`pdk-monitor.ai.chat.history`). Clear conversation wipes it but keeps settings + per-BB caches. ⬇ Export .md downloads the full transcript including every tool call and input.

Cross-provider: Anthropic uses native tool-use; OpenAI-compatible endpoints use OpenAI's tools schema. The dashboard translates between the two message shapes so you can switch providers without clearing history.

25.5 Contributor Excel report

The Contributors modal footer now has a Download Excel report button visible in every mode (read-only, user-admin, admin). It generates `contributors-statistics-YYYY-MM-DD.xlsx` with five sheets:

Summary — one row per contributor with total BBs, variants, platforms touched, file counts broken down by section (layout / S-matrix / tests / fab / scripts / docs / EDA / other), first + last contribution dates, source (Manual / Auto-detected), affiliation. Sorted by total activity. Final TOTAL row across everyone.

Per-platform — pivot matrix: rows are contributors, columns are every platform on the dashboard, cells are BB counts, with row + column totals.

BBs — flat list of every BB with owner, platform, BB-type, status, category, last-updated date, ID. Variants flagged separately.

Files — flat list of every uploaded file with owner, section, filename, size, last-modified date, parent BB id. Final TOTAL row.

About — workbook metadata: generation timestamp, counts of manual + auto-detected identities, instructions for re-export.

The walker recurses through subBlocks so variants are attributed correctly. Manual contributor records (Name + Affiliation from the modal) take precedence over auto-detected identities for the display name in the report.

25.6 Cost meter and safety

Anthropic and OpenAI both return usage objects after every call; the dashboard parses both and tracks input + output tokens per month under `pdk-monitor.ai.usage`. The AI

Settings modal shows a progress bar. When the cap is reached every button disables with a tooltip — raise the cap or wait for the calendar month to roll over.

Rough cost expectations: a per-BB summary is \$0.01-\$0.05 on Sonnet, \$0.05-\$0.25 on Opus, under \$0.005 on Haiku 4.5. Caching is content-keyed, so a 30-BB dashboard refresh after all summaries have been generated once is essentially free.

API keys live in localStorage only. The publish payload strip in pruneOrphanRepoFiles + the binary-fields purge in the publish helper both already remove `pdk-monitor.ai.*` keys from anything that leaves the browser, so accidentally publishing your API key is not possible.

25.7 Modern AI logo

The dashboard's AI entry-points now use a CSS-only conic-gradient circle (purple → pink → orange → cyan → purple) with white 'AI' lettering inside, an 8-second rotating ring, and a 3.6-second purple-to-cyan glow pulse. Three sizes (.sm 18 px, default 22 px, .lg 30 px) are exposed so the same mark can show up at any size in future UI work.

26. Revision 6 — sync fidelity, moveable modals, cross-platform BB move

Revision 6 makes the dashboard noticeably more fluid in day-to-day use. The sync flow now mirrors GitHub reality, every modal is moveable + minimizable so long-running operations can be tucked away while you keep browsing, and a new Move to platform action lets you reorganise BBs across Technology Platforms without manually re-creating them.

26.1 Sync now mirrors the GitHub remote

When you delete a file from your GitHub repo and click  Sync, the local copy of the file used to linger because the old sync only wiped the canonical 'Files' folder for the section. Revision 6 fixes this with a pre-wipe step:

Before the URLs are replayed, every folder of each (BB, section) pair touched by the sync is cleared — `testGroups[].tests`, `smatrixGroups[].datasets`, `fabTabs[].tests`, `scripts`. The sync log reports how many stale files were dropped.

Each URL is then replayed in append mode (no more `replace:true`) so multiple URLs configured for the same section UNION correctly instead of clobbering each other.

End state of the local view = the union of every configured URL's current contents at this moment. Files deleted on the remote drop out; files added on the remote appear; files moved between folders land wherever the section-fetch decides.

26.2 Live file-count badge on the sync buttons

Every sync run now tracks how many files were actually fetched and how many URLs returned nothing (= the files that were removed remotely). Stats are stored on the right scope:

Per-BB sync $\rightarrow$ e.lastSyncStats

Per-platform sync $\rightarrow$ g.lastSyncStats

Whole-dashboard sync $\rightarrow$ MODEL.lastGlobalSyncStats

Each record carries { ranAt, urlsTotal, filesFound, urlsEmpty, deadUrls[] }. The buttons re-render from these stats on every render() call:

Platform group head: $\cup$ Sync $\cdot$ X files $\cdot$ Y removed (Y in red if > 0)

Header $\cup$ Sync all: same suffix

Per-BB $\cup$ Sync (N): N is the number of WORKING URLs (urlsTotal – urlsEmpty). Before any sync has run it falls back to fetchHistory.length so the button is never empty. Tooltip carries the verbose breakdown.

26.3 Dead-URL pruning

If the latest sync detected URLs that returned nothing, a Remove N dead URLs from history button appears in the sync modal footer. Click it and the dashboard prunes those URLs out of every affected BB's fetchHistory so future syncs no longer waste time on them. Confirmation dialog explains exactly what will be removed; nothing is touched on GitHub.

The sync log itself reports each URL's contribution inline: +5 files from this URL, or $\triangle 0$ files — this URL is empty / removed on the remote — so you can see which URLs are dead before you decide whether to prune.

26.4 Moveable + minimizable modals

Every popup in the dashboard — Sync, Cross-check, AI Insight, Chat, Prepare-for-PDK, Manage lists, Contributors, Distribution, BB edit, every fetch — now has a draggable header and a minimize button.

:: grip glyph at the left of the title makes the drag affordance discoverable. The entire header bar is the drag handle (just not the buttons inside it). The cursor turns to a grabbing hand during drag.

The modal is clamped so the title bar can never disappear off-screen — you can always grab it again.

— button between the title and the $\times$ is the minimize toggle. Click to collapse the modal to a 360×48 pixel pill at the bottom-right with an accent-coloured border. Click the pill or its $\swarrow$ button to restore.

While the modal is open or minimized, the dashboard behind stays fully interactive. Click BBs, browse tabs, edit unrelated panels — the modal stays where you put it. Long syncs / AI calls / fetches no longer block browsing.

Every fresh openModal() call resets position to centred and un-minimises, so opening a new dialog doesn't inherit the last one's location.

26.5 ? Move BB to another Technology Platform

A new 

Move to platform button

and Edit. Available to admin and to the BB's owner (user-admin who created it); hidden in read-only mode and on docs-kind entities.

Clicking it opens a draggable + minimizable modal with:

Move from card — current platform name plus material/thickness for orientation.

Move to dropdown — every other Technology Platform on the dashboard, each labelled with name, material/thickness, current BB count, and parent tab.

Heads-up panel — three points: GitHub fetch URLs may need re-pointing after the move, same-named BBs on the destination don't auto-merge (use the duplicate resolver afterwards if you want), and already-published users/<id>/ files stay where they are until you next publish from the new home.

Confirm with  Move BB and the dashboard splices the entity out of the source group's entities[] array, appends it to the destination group's, refreshes e.platform, persists, switches the visible tab to the destination and re-opens the BB Details panel on the moved BB. Every file the BB carries — GDS / OASIS layout, S-matrix datasets, test groups, fab folders, scripts, documents, EDA links, variants (sub-blocks), fetchHistory, lastSyncStats, ownership, AI cache — travels with the entity because each BB is a single object reference; the splice is enough.

If there's only one platform on the dashboard, clicking the button shows a toast explaining you need at least one more platform via + Add tab or Manage → Platforms before you can move anything.

27. Version history

v1.0.0 (rev. 6, 28 May 2026) — Sync now mirrors the GitHub remote with a pre-wipe before replay; live file-count badge on every sync button (X files · Y removed); per-BB sync button simplified to  Sync (N) of working links only; dead-URL pruning button when any URL returned nothing; every modal is now draggable by header and minimizable to a corner pill; new  Move BB to another Technology Platform action that splices a BB across platforms with every file, variant, history record and AI cache travelling along.

v1.0.0 (rev. 6, 27 May 2026) — AI Insight end-to-end: configurable endpoint (Anthropic / OpenAI / OpenRouter / Ollama / vLLM / LM Studio) with monthly token cap and three egress modes, Test connection + Reveal key + key meta line, per-BB AI tab with deterministic metric extraction + citation discipline + content-hash cache, platform + dashboard executive summaries, multi-turn chat agent with six read-only tools (dashboard / platforms / BBs / contributors / metrics / search / summarise), Contributor Excel report in the Contributors modal (5 sheets), modern conic-gradient AI logo replacing the .

v1.0.0 (rev. 6, 27 May 2026) — Admin safety gate against users/ publishes, cross-check analysis suite with three donut charts + Excel report + selective per-BB zip download, variants as first-class category (header chip + Excel sheet + donut), admin-editable footer, Clear cache in every role, Prepare-for-PDK opened to Admin, scroll fixed across every BB tab, tab-named externalize subfolders, instant BB-card refresh after upload, platform column width tuning, custom BB type labels everywhere, collective-contributor permissions, lazy-hydrate downloads with correct MIME.

v1.0.0 rev. 3 — 27 May 2026 — Layout section accepts GDS + OASIS (.oas / .oasis) as essential layout formats, SVG kept as preview fallback. Prepare-for-PDK polish: per-section file multi-select, ✂ Direct dual publish buttons with cooldowns (1/hour PR; progressive direct commit allowing 2/min and 5/10 min), no README on commit, no SVG upload. Editable target URL on Commit-tests-to-repo. Scripts / S-matrix GitHub fetch accept every file type. Lazy-hydrate on ⬇ Download — cross-user files come down as the original bytes with the correct MIME, not a JSON sidecar. bbTypeLabel() resolver — custom Manage-list BB types show their proper label everywhere. Inner folder scroll (~3 items) + outer panel scroll work together; × reachable on every file row. User-admin can edit BBs created by other users.

- **v1.0.0 rev. 2 — 26 May 2026** — Final post-release polish: selective fetch picker, read-only resize + cache-clear, per-row Skip/Merge-in in cross-platform resolver, BB-level zip download, auto-YAML on GDS upload, material-neutral photodetector sketch, ownership backfill from who.
- **v1.0.0 — 26 May 2026** — Initial public release.